The Value of Knowing a Demand Curve: Bounds on Regret for Online Posted-Price Auctions

EXTENDED ABSTRACT

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Abstract

We consider price-setting algorithms for a simple market in which a seller has an unlimited supply of identical copies of some good, and interacts sequentially with a pool of n buyers, each of whom wants at most one copy of the good. In each transaction, the seller offers a price between 0 and 1, and the buyer decides whether or not to buy, by comparing the offered price to his privately-held valuation for the good. The price offered to a given buyer may be influenced by the outcomes of prior transactions, but each individual buyer participates only once.

In this setting, what is the value of knowing the demand curve? In other words, how much revenue can an uninformed seller expect to obtain, relative to a seller with prior information about the buyers' valuations? The answer depends on how the buyers' valuations are modeled. We analyze three cases — identical, random, and worst-case valuations — in each case deriving upper and lower bounds which match within a sublogarithmic factor.

1 Introduction

Imagine that you are operating a website which sells many identical copies of a good using the most familiar transaction model: each site visitor is offered a price for the good, and the visitor may either buy the good at the offered price or go away. (This is known as an online posted-price mechanism.) To maximize revenue, it is necessary to find the optimal price to offer for the good, and there are two ways to do this: pay a marketing research firm to determine the demand curve, or learn the optimal price gradually through experimentation with actual buyers. We are interested in the question, "What is the value of knowing the demand curve?" In other words, we wish to determine the difference in expected revenue between an informed seller and one who sets prices according to the optimal online strategy. (This quantity is hereafter referred to as the *regret* of the online pricing strategy.) One can envision several variants of this problem, depending on how one models the buyers' valuations and how one defines the "informed seller." We will study three valuation models.

Identical: All buyers' valuations are equal to a single price $p \in [0, 1]$.

Random: Buyers' valuations are independent random samples from a fixed probability distribution on $[0, 1]$.

Worst-case: The model makes no assumptions about the buyers' valuations. They may be chosen by an adversary who is oblivious to the random choices made by the seller's pricing strategy.

In the case of identical and worst-case valuations, it is customary to model the informed seller as one who knows exactly how much each buyer is willing to pay, but is constrained to set a fixed price for the good, and chooses the optimal price subject to this constraint. In the case of random valuations we may adopt the same definition, but it is more natural to define the informed seller to be one who knows the probability distribution from which buyers' valuations are sampled (but not the samples themselves) and sets the best price given this knowledge. We refer to the regret in the former case as *ex post* regret, and in the latter case as *ex ante* regret. Note that the ex ante regret comes closest to the meaning of the original question, "What is the value of knowing the demand curve?" We will prove a lower bound on ex ante regret combined with a nearly-matching upper bound on ex post regret; note that this is the strongest pair of bounds one could hope for, since ex post regret always exceeds ex ante regret.

We will always assume that prices are constrained to lie between 0 and 1, and that there are n buyers, where the value of n is known to the seller. The latter assumption

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only makes it harder to prove lower bounds, and our upper bounds will actually hold regardless of whether the seller has foreknowledge of n. Our results are summarized in the following three theorems. In all of them, the term "pricing strategy" refers to a randomized online algorithm for choosing offer prices, unless noted otherwise.

Theorem 1.1. *Assuming identical valuations, there is a deterministic pricing strategy achieving regret $O(\log\log n)$. No pricing strategy can achieve regret $o(\log\log n)$.*

Theorem 1.2. *Assuming random valuations, there is a pricing strategy achieving ex post regret $O(\sqrt{n\log n})$, under the hypothesis that the function*

$$f(x) = x \cdot \Pr(\textit{buyer's valuation is} \geq x)$$

has a unique global maximum x^* in the interior of $[0,1]$, and that $f''(x^*) < 0$. No pricing strategy can achieve ex ante regret $o(\sqrt{n})$, even under the same hypothesis on the distribution of valuations.

Theorem 1.3. *Assuming worst-case valuations, there is a pricing strategy achieving regret $O((n^{2/3}(\log n)^{1/3})$. No pricing strategy can achieve regret $o(n^{2/3})$.*

The lower bound in the random-valuation model is the most difficult of the results stated above, and we believe it is this paper's main contribution. It improves the best previously-known lower bound of $\Omega(\log n)$ by an exponential amount, and our proof introduces novel techniques which we believe may be applicable elsewhere. Moreover, our lower bound does not rely on constructing a contrived demand curve to defeat a given pricing strategy. Rather, we will show that for *any* family $\mathcal{D}$ of demand curves satisfying some reasonably generic axioms, and for any randomized pricing strategy, the probability of achieving expected regret $o(\sqrt{n})$, when the demand curve is chosen randomly from $\mathcal{D}$, is zero. To achieve this, we follow an intuitively natural line of argument: we define a quantitative measure of "knowledge," prove that a pricing strategy must accumulate a non-negligible amount of knowledge about the demand curve in order to avoid having high regret, and place a lower bound on the cost of acquiring this amount of knowledge.

1.1 Related work

There has been much recent activity applying notions from the theory of algorithms to the analysis of auction mechanisms. While much of this work focuses on combinatorial auctions — a subject not touched on here — there has also been a considerable amount of work on auction mechanisms for selling identical individual items, the setting considered in this paper. Competitive analysis of such auctions was introduced in [7], [8], where the authors consider strategyproof mechanisms for off-line auctions, i.e. those in which all buyers reveal their valuations before any goods are sold. On-line auctions were considered in [5], [6], in the posted-price setting considered here as well as the setting where buyers reveal their valuations but are charged a price which depends only on the information revealed by prior buyers. In the latter paper, techniques from machine learning theory are applied to yield a $(1+\varepsilon)$-competitive on-line mechanism (for any $\varepsilon > 0$) under the hypothesis that the optimal single-price auction achieves revenue $\Omega(h\log h\log\log h)$, where $[1,h]$ is the interval to which the buyers' valuations belong. In Section 4.1, we use their algorithm (with a very minor technical modification) to achieve expected regret $O(n^{2/3}(\log n)^{1/3})$ assuming worst-case valuations. On-line multi-unit auctions (in which buyers may bid for multiple copies of the item) are considered in [4] and [10]; the latter paper exactly determines the optimal competitive ratio, as a function of the ratio between the highest and lowest unit prices offered. An interesting hybrid of the off-line and on-line settings is considered by Hartline in [9]. In that paper, the mechanism interacts with the set of buyers in two rounds, where prices in the second round are influenced by the preferences revealed by buyers in the first round.

The preceding papers have all adopted the worst-case model for buyers' valuations, as is customary in the computer science literature. The traditional approach in economics (e.g. [12]) is to assume that buyers' valuations are i.i.d. samples from a *known* probability distribution. Our random-valuations model occupies a middle ground between these two sets of assumptions, in that the i.i.d. hypothesis is preserved but the probability distribution (i.e. the demand curve) is unknown to the seller. The same set of hypotheses is made by Ilya Segal in [13], a paper which considers strategyproof off-line multi-unit auction mechanisms. Though the paper is primarily concerned with off-line mechanisms, Segal briefly mentions online posted-price mechanisms (which he calls "experimentation mechanisms") noting, "The optimal experimentation mechanism would be very difficult to characterize [in the case of general demand curves]. Intuitively, it appears that its convergence rate may be slower [than that of the optimal off-line mechanism] because the early purchases at prices that are far from p^* will prove useless for fine-tuning the price around p^*." This intuition is confirmed by the lower bound we prove in Section 3.2.

Our work is also closely tied to the literature on the so-called "multi-armed bandit problem," in which a gambler in a casino with K slot machines must decide which machine to play in a sequence of n trials, basing his decisions on the payoffs observed in prior trials. As in our auction problem, the regret is defined as the difference between the gambler's expected net payoff and the net payoff obtained

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from the best single action (slot machine) over the sequence of n trials. In their pioneering work on this problem, Lai and Robbins [11] assumed that for each action, the payoffs on each of the n trials are i.i.d. random variables, but the distribution of the payoff variable varies from one action to another. (As before, there is a distinction between ex ante and ex post regret in this model.) They exhibited an algorithm achieving ex ante regret $O(\log n)$ as $n \to \infty$, and proved that this is best possible, up to constant factors. Auer et al [2] sharpened this analysis to obtain explicit regret bounds which hold for finite n.

If we view each potential offer price in $[0, 1]$ as a slot machine with random payoff, then our on-line posted-price auction problem (in the random-valuations model) becomes a special case of the "continuum-armed bandit problem", in which the finite set of slot machines is replaced by an uncountable set indexed by a real parameter t, with the expected reward depending continuously on t. One might expect that the regret still grows logarithmically in n, by analogy with the finite-armed case. However the best known upper bound is due to Agrawal [1], who describes an algorithm achieving regret $O(n^{3/4+\varepsilon})$ in the case that the expected reward is a differentiable function of t. This upper bound is weaker than our bound of $O(\sqrt{n \log n})$ in the random-valuations model, as might be expected because our auction problem is a very specific case of the general continuum-armed bandit problem. However, our lower bound of $\Omega(\sqrt{n})$ for the random-valuations model immediately implies the same lower bound for the continuum-armed bandit problem. Previously the best-known lower bound was $\Omega(\log n)$, coming from the finite-armed bandit analysis by Lai and Robbins [11]. Our paper is thus the first demonstration of an exponential separation between the expected regret when the set of possible actions is finite versus when it is infinite.

In [3], the multi-armed bandit problem is studied from an adversarial perspective, parallel to our worst-case valuation model. The authors present an algorithm achieving expected regret $O(\sqrt{nK \log K})$ and a nearly-matching lower bound of $\Omega(\sqrt{nK})$ for this problem. Their algorithm forms the basis for the on-line posted-price auction algorithms in [6] and in Section 4.1 of this paper, and our lower-bound proof in the worst-case model (Theorem 4.3) is an adaptation of their lower-bound proof. However, the resemblance between their lower bound and our *random-valuations* lower bound of $\Omega(\sqrt{n})$ is misleading. They assume a stronger adversary and prove a lower bound on ex post regret. Our random-valuations lower bound refers to ex ante regret, which is not even defined in the adversarial setting. (The lower bound of $\Omega(\sqrt{n})$ for *ex post* regret with random valuations is much easier to prove.) The only lower bound on ex ante regret existing in the multi-armed bandit literature was the Lai-Robbins lower bound of $\Omega(\log n)$.

2 Identical valuations

2.1 Upper bound

When all buyers have the same valuation $p \in [0, 1]$, every response gives the seller perfect information about a lower or upper bound on p, depending on whether the buyer's response was to accept or to reject the price offered. A pricing strategy $\mathcal{S}$ which achieves regret $O(\log \log n)$ may be described as follows. The strategy keeps track of a *feasible interval* $[a, b]$, initialized to $[0, 1]$, and a *precision parameter* ε, initialized to $1/2$. In a given phase of the algorithm, the seller offers the prices $a, a+\varepsilon, a+2\varepsilon, \ldots$ until one of them is rejected. If $a + k\varepsilon$ was the last offer accepted in this phase, then $[a + k\varepsilon, a + (k + 1)\varepsilon]$ becomes the new feasible interval, and the new precision parameter is ε^2. This process continues until the length of the feasible interval is less than $1/n$; then the seller offers a price of a to all remaining buyers.

Theorem 2.1. *The regret of strategy $\mathcal{S}$ is bounded above by* $2\lceil \log_2 \log_2 n \rceil + 4$.

Proof. The number of phases is equal to the number of iterations of repeated squaring necessary to get from $1/2$ to $1/n$, i.e. $\lceil \log_2 \log_2 n \rceil + 1$. Let p denote the valuation shared by all buyers. The seller accrues regret for two reasons:

- Items are sold at a price $q < p$, accruing regret $p - q$.
- Buyers decline items, accruing regret p.

At most one item is declined per phase, incurring at most one unit of regret, so the declined offers contribute at most $\lceil \log_2 \log_2 n \rceil + 1$ to the total regret.

In each phase except the first and the last, the length $b-a$ of the feasible interval is $\sqrt{\varepsilon}$ (i.e. it is the value of ε from the previous phase), and the set of offer prices carves up the feasible interval into subintervals of length ε. There are $1/\sqrt{\varepsilon}$ such subintervals, so there are at most $1/\sqrt{\varepsilon}$ offers made during this phase. Each time one of them is accepted, this contributes at most $b - a = \sqrt{\varepsilon}$ to the total regret. Thus, the total regret contribution from accepted offers in this phase is less than or equal to $(1/\sqrt{\varepsilon}) \cdot \sqrt{\varepsilon} = 1$. The first phase is exceptional since the feasible interval is longer than $\sqrt{\varepsilon}$. The accepted offers in phase 1 contribute at most 2 to the total regret. Summing over all phases, the total regret contribution from accepted offers is $\leq \lceil \log_2 \log_2 n \rceil + 3$.

In the final phase, the length of the feasible interval is at most $1/n$, and each offer is accepted. There are at most n such offers, so they contribute at most 1 to the total regret. □

Remark 2.2. If the seller does not have foreknowledge of n, it is still possible to achieve regret $O(\log \log n)$ by modifying the strategy. At the start of a phase in which the feasible interval is $[a, b]$, the seller offers price a to the next

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$\lfloor 1/(b-a) \rfloor$ buyers. This raises the regret per phase from 2 to 3, but ensures that the number of phases does not exceed $\lceil \log_2 \log_2 n \rceil + 1$.

2.2 Lower bound

Theorem 2.3. *If $\mathcal{S}$ is any randomized pricing strategy, and p is randomly sampled from the uniform distribution on $[0,1]$, the expected regret of $\mathcal{S}$ when the buyers' valuations are p is $\Omega(\log \log n)$.*

Proof. It suffices to prove the lower bound for a deterministic pricing strategy $\mathcal{S}$, since any randomized pricing strategy is a probability distribution over deterministic ones. At any stage of the game, let a denote the highest price that has yet been accepted, and b the lowest price that has yet been declined; thus $p \in [a,b]$. As before, we will refer to this interval as the *feasible interval*. It is counterproductive to offer a price less than a or greater than b, so we may assume that the pricing strategy works as follows: it offers an ascending sequence of prices until one of them is declined; it then limits its search to the new feasible interval, offering an ascending sequence of prices in this interval until one of them is declined, and so forth.

Divide the pool of buyers into phases (starting with phase 0) as follows: phase k begins immediately after the end of phase $k-1$, and ends after an addtional $2^{2^k}-1$ buyers, or after the first rejected offer following phase $k-1$, whichever comes earlier. The number of phases is $\Omega(\log \log n)$, so it suffices to prove that the expected regret in each phase is $\Omega(1)$. This is established by the following three claims. □

Claim 2.4. *Let $\mathcal{I}_k$ denote the set of possible feasible intervals at the start of phase k. The cardinality of $\mathcal{I}_k$ is at most 2^{2^k}, and the intervals in $\mathcal{I}_k$ have disjoint interiors.*

Proof. The proof is by induction on k. The base case $k=0$ is trivial. Now assume the claim is true for a particular value of k, and let Let $I_k = [a_k, b_k]$ be the feasible interval at the start of phase k. Let $x_1 \le x_2 \le \cdots \le x_j$ denote the ascending sequence of prices that $\mathcal{S}$ will offer during phase k if all offers are accepted. (Here $j = 2^{2^k}-1$.) Then the feasible interval at the start of phase $k+1$ will be one of the subintervals $[a_k, x_1], [x_1, x_2], [x_2, x_3], \ldots, [x_{j-1}, x_j], [x_j, b_k]$. There are at most $j = 2^{2^k}$ such subintervals, and at most 2^{2^k} possible choices for I_k (by the induction hypothesis), hence there are at most $2^{2^{k+1}}$ elements of $\mathcal{I}_{k+1}$. Moreover, they all have disjoint interiors because we have simply taken the intervals in $\mathcal{I}_k$ and partitioned them into subintervals. □

Claim 2.5. *Let $|I|$ denote the length of an interval I. With probability at least 3/4, $|I_k| \ge \frac{1}{4} \cdot 2^{-2^k}$.*

Proof. The expectation of $1/|I_k|$ may be computed as follows:

$$\mathbf{E}(1/|I_k|) = \sum_{I \in \mathcal{I}_k} \Pr(p \in I)(1/|I|) = \sum_{I \in \mathcal{I}_k} |I|/|I| \le 2^{2^k},$$

where the last inequality follows from Claim 2.4. Now use Markov's Inequality:

$$\Pr(|I_k| < \frac{1}{4} \cdot 2^{-2^k}) = \Pr(1/|I_k| > 4 \cdot 2^{2^k}) < 1/4.$$

□

Claim 2.6. *The expected regret in phase k is at least $\frac{1}{128}$.*

Proof. By Claim 2.4, with probability 1, p belongs to a unique interval $I_k = [a,b] \in \mathcal{I}_k$. Let $\mathcal{E}_k$ denote the event that $b \ge 1/4$ and $|I_k| \ge \frac{1}{4} \cdot 2^{-2^k}$. This is the intersection of two events, each having probability $\ge 3/4$, so $\Pr(\mathcal{E}_k) \ge 1/2$. It suffices to show that the expected regret in phase k, conditional on $\mathcal{E}_k$, is at least 1/64. So from now on, assume that $p \ge 1/4$ and $|I_k| \ge \frac{1}{4} \cdot 2^{-2^k}$. Also note that, conditional on the events $\mathcal{E}_k$ and $p \in I_k$, p is uniformly distributed in I_k.

Let $m = (a+b)/2$. As before, let $j = 2^{2^k}-1$ and let $x_1 \le x_2 \le \cdots \le x_j$ denote the ascending sequence of prices which $\mathcal{S}$ would offer in phase k if no offers were rejected. We distinguish two cases:

Case 1: $x_j \ge m$**.** With probability at least 1/2, $p < m$ and the phase ends in a rejected offer, incurring a regret of p, whose conditional expectation $\frac{a+m}{2}$ is at least 1/16. Thus the expected regret in this case is at least 1/32.

Case 2: $x_j < m$**.** The event $\{p > m\}$ occurs with probability 1/2, and conditional on this event the expectation of $p - m$ is $|I_k|/4 \ge 2^{-2^k}/16$. Thus with probability at least 1/2, there will be $2^{2^k}-1$ accepted offers, each contributing $2^{-2^k}/16$ to the expected regret, for a total of $(1/2)(2^{2^k}-1)(2^{-2^k})/16 \ge 1/64$.

□

3 Random valuations

3.1 Preliminaries

In this section we will consider the case where each buyer's valuation v is an independent random sample from a fixed but unknown probability distribution on $[0,1]$. It is customary to describe this probability distribution in terms of its *demand curve*

$$D(x) = \Pr(v \ge x).$$

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Given foreknowledge of the demand curve, but not of the individual buyers' valuations, it is easy to see what the optimal pricing strategy $\mathcal{S}^*$ would be. The expected revenue obtained from setting price x is $xD(x)$. Since buyers' valuations are independent and the demand curve is known, the individual buyers' responses provide no useful information about future buyers' valuations. The best strategy is thus to compute

$$x^* = \arg\max_{x\in[0,1]} xD(x)$$

and to offer this price to every buyer.

As was discussed in the introduction, a pricing strategy $\mathcal{S}$ may be judged according to its *ex ante* regret $\rho(\mathcal{S}^*) - \rho(\mathcal{S})$ or its *ex post* regret $\rho(\mathcal{S}^{\mathrm{opt}}) - \rho(\mathcal{S})$. Our lower bound on ex ante regret implies the same bound for ex post regret, and vice-versa in the case of the upper bound.

A deterministic pricing strategy can be specified by a sequence of rooted binary trees $T_1, T_2, \ldots$, where the n-th tree specifies the decision tree to be applied by the seller when interacting with a population of n buyers. (Thus T_n is a complete binary tree of depth n.) We will use a to denote a generic internal node of such a decision tree, and ℓ to denote a generic leaf. The relation $a \prec b$ will denote that b is a descendant of a; here b may be a leaf or another internal node.

The internal nodes of the tree are labeled with numbers $x_a \in [0,1]$ denoting the price offered by the seller at node a, and random variables $v_a \in [0,1]$ denoting the valuation of the buyer with whom the seller interacts at that node. The buyer's choice is represented by a random variable

$$\chi_a = \begin{cases} 1 & \text{if } v_a \geq x_a \\ 0 & \text{if } v_a < x_a. \end{cases}$$

In other words, χ_a is 1 if the buyer accepts the price offered, 0 otherwise.

The tree T_n specifies a pricing strategy as follows. The seller starts at the root r of the tree and offers the first buyer price x_r. The seller moves from this node to its left or right child depending on whether the buyer declines or accepts the offer, and repeats this process until reaching a leaf which represents the outcome of the auction. A probability distribution on the buyers' valuations v_a induces a probability distribution on outcomes ℓ. We will use $p_D(\ell)$ to denote the probability assigned to ℓ under the valuation distribution represented by demand curve D. For an internal node a, $p_D(a)$ denotes the probability that the outcome leaf is a descendant of a.

3.2 Lower bound

3.2.1 A family of random demand curves

The demand curves D appearing in our lower bound will be random samples from a space $\mathcal{D}$ of possible demand curves. In this section we single out a particular random demand-curve model, and we enumerate the properties which will be relevant in establishing the lower bound (Theorem 3.9). The choice of a particular random demand-curve model is done here for ease of exposition, and not because of a lack of generality in the lower bound itself. In Section 3.2.6 we will indicate that Theorem 3.9 applies to much broader classes $\mathcal{D}$ of demand curves. In particular we believe that it encompasses demand-curve models which are realistic enough to be of interest in actual economics and e-commerce applications.

For now, however, $\mathcal{D}$ denotes the one-parameter family of demand curves $\{D_t : 0.3 \leq t \leq 0.4\}$ defined as follows. Let

$$\tilde{D}_t(x) = \max\left\{1 - 2x, \frac{2t - x}{7t^2}, \frac{1-x}{2}\right\}.$$

In other words, the graph of $\tilde{D}_t$ consists of three line segments: the middle segment is tangent to the curve $xy = 1/7$ at the point $(t, 1/7t)$, while the left and right segments belong to lines which lie below that curve and are independent of t. Now we obtain D_t by smoothing $\tilde{D}_t$. Specifically, let $b(x)$ be a non-negative, even C^∞ function supported on the interval $[-0.01, 0.01]$ and satisfying $\int_{-0.01}^{0.01} b(x)\,dx = 1$. Define D_t by convolving $\tilde{D}_t$ with b. We will equip $\mathcal{D} = \{D_t : 0.3 \leq t \leq 0.4\}$ with a probability measure by specifying that t is uniformly distributed in $[0.3, 0.4]$.

Let $x_t^* = \arg\max_{x\in[0,1]} xD_t(x)$. It is an exercise to compute that $x_t^* = t$. (With $\tilde{D}_t$ in place of D_t this would be trivial. Now $D_t(x) = \tilde{D}_t(x)$ unless x is within 0.01 of one of the two points where $\tilde{D}_t'$ is discontinuous, and these two points are far from maximizing $x\tilde{D}_t(x)$, so $xD_t(x)$ is also maximized at $x = t$.)

We now enumerate the properties of this class of demand curves $\mathcal{D}$ that will be required in proving the lower bound. We have omitted the proofs, which are exercises in basic calculus. The meaning of the properties is summarized, at an intuitive level, at the start of the next section.

Lemma 3.1. *There exist constants $\alpha, \beta > 0$ and $\gamma < \infty$ such that for all $D = D_{t_0} \in D$ and $x \in [0,1]$:*

1. $d/dt(x_t^*)|_{t=t_0} > \alpha$;
2. $x^*D(x^*) - xD(x) > \beta(x^* - x)^2$;
3. $|\dot{D}(x)/D(x)| < \gamma|x^* - x|$ *and* $|\dot{D}(x)/(1 - D(x))| < \gamma|x^* - x|$.
4. $|D^{(k)}(x)/D(x)| < \gamma$ *and* $|D^{(k)}(x)/(1 - D(x))| < \gamma$, *for* $k = 2, 3, 4$.

Here x^* denotes $x_{t_0}^*$, $D^{(k)}(x)$ denotes the k-th t-derivative of $D_t(x)$ at $t = t_0$, and $\dot{D}(x)$ denotes $D^{(1)}(x)$.

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3.2.2 High-level description of the proof

The proof of the lower bound on regret is based on the following intuition. If there is uncertainty about the demand curve, then no single price can achieve a low expected regret for all demand curves. The family of demand curves exhibited above is parametrized by a single parameter t, and we will see that if the uncertainty about t is on the order of ε then the regret per buyer is $\Omega(\varepsilon^2)$. (This statement will be made precise in Lemma 3.7 below.) So to avoid accumulating $\Omega(\sqrt{n})$ regret on the last $\Omega(n)$ buyers, the pricing strategy must ensure that it reduces the uncertainty to $O(n^{-1/4})$ during its interactions with the initial $O(n)$ buyers. However — and this is the crux of the proof — we will show that offering prices far from x^* is much more informative than offering prices near x^*, so there is a quantifiable cost to reducing the uncertainty in t. In particular, reducing the uncertainty to $O(n^{-1/4})$ costs $\Omega(\sqrt{n})$ in terms of expected regret.

To make these ideas precise, we will introduce a notion of "knowledge" which quantifies the seller's ability to distinguish the actual demand curve from nearby ones based on the information obtained from past transactions, and a notion of "conditional regret" whose expectation is equal to the pricing strategy's expected regret. We will show that the ratio of conditional regret to knowledge is bounded below, so that the strategy cannot accumulate $\Omega(\sqrt{n})$ knowledge without accumulating $\Omega(\sqrt{n})$ regret. Finally, we will show that when the expected knowledge is less than a small constant multiple of $\sqrt{n}$, there is so much uncertainty about the true demand curve that the expected regret is $\Omega(\sqrt{n})$ with high probability (taken over the probability measure on demand curves).

3.2.3 Definition of knowledge

In the following definitions, "log" denotes the natural logarithm function. T denotes a finite binary tree, labeled with a pricing strategy as explained in Section 3.1. When f is a function defined on leaves of T, we will use the notation $\mathbf{E}_D f(\ell)$ to denote the expectation of f with respect to the probability distribution p_D on leaves, i.e.

$$\mathbf{E}_D f(\ell) = \sum_{\ell \in T} p_D(\ell) f(\ell).$$

For a given demand curve $D = D_{t_0}$, we define the *infinitesimal relative entropy* of a leaf $\ell \in T$ by

$$IRE_D(\ell) = \frac{d}{dt}\left(-\log p_{D_t}(\ell)\right)|_{t=t_0}.$$

Those familiar with the notion of relative entropy from information theory may recognize $IRE_D(\ell)$ as the t-derivative of ℓ's contribution to the weighted sum defining $RE(D\|D_t)$. We define the *knowledge* of ℓ as the square of the infinitesimal relative entropy:

$$K_D(\ell) = IRE_D(\ell)^2.$$

Note that when $K_D(\ell)$ is small, the absolute value of $IRE_D(\ell)$ is also small, i.e. $p_{D_t}(\ell)$ is slowly-varying as a function of t when t is near t_0. This means that the outcome ℓ is just about as likely to be observed if the demand curve is D or if it is a nearby demand curve D', so a seller observing outcome ℓ can not confidently distinguish whether the demand curve is D or D'. This corresponds intuitively to the notion that outcome ℓ provides the seller with only a small amount of knowledge about D, and explains the motivation behind our terminology.

An important feature of $IRE_D(\ell)$ is that it may be expressed as a sum of terms coming from the ancestors a of ℓ. Let

$$\begin{aligned} ire_D(x_a, \chi_a) &= \begin{cases} \frac{d}{dt}(\log D(x_a)) & \text{if } \chi_a = 1 \\ \frac{d}{dt}(\log(1 - D(x_a))) & \text{if } \chi_a = 0 \end{cases} \\ &= \begin{cases} \dot{D}(x_a)/D(x_a) & \text{if } \chi_a = 1 \\ -\dot{D}(x_a)/(1 - D(x_a)) & \text{if } \chi_a = 0 \end{cases} \end{aligned}$$

Then

$$IRE_D(\ell) = \sum_{a \prec \ell} ire_D(x_a, \chi_a).$$

3.2.4 Definition of conditional regret

For a given D, the conditional regret $R_D(\ell)$ may be informally defined as follows. At the end of the auction, if the demand curve D were revealed to the seller and then she were required to repeat the same sequence of offered prices $\{x_a : a \prec \ell\}$ to a new, independent random population of buyers whose valuations are distributed according to D, then $R_D(\ell)$ is the expected regret incurred by the seller during this second round of selling. Formally, $R_D(\ell)$ is defined as follows. Let

$$r_D(x) = x^* D(x^*) - x D(x),$$

where $x^* = \arg\max_{x \in [0,1]}\{x D(x)\}$ as always. Note that if two different sellers offer prices x^*, x, respectively, to a buyer whose valuation is distributed according to D, then $r_D(x)$ is the difference in their expected revenues. Now let

$$R_D(\ell) = \sum_{a \prec \ell} r_D(x_a).$$

Although $R_D(\ell)$ is *not* equal to the seller's actual regret conditional on outcome ℓ, it is a useful invariant because its expectation $\mathbf{E}_D R_D(\ell)$ is equal to the *ex ante* regret of $\mathcal{S}$. This fact is not obvious, because the distribution of the

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actual buyers' valuations, conditioned on their responses to the prices they were offered, is very different from the distribution of n new independent buyers. Yet the expected difference in revenue between the two strategies (i.e. the regret) is the same for both populations of buyers, as is proved in the following lemma. (As always, $\rho(\cdot)$ denotes expected revenue of a pricing strategy under the distribution p_D.)

Lemma 3.2. *If the buyers' valuations are independent random samples from the distribution specified by D, then $\rho(\mathcal{S}^*) - \rho(\mathcal{S}) = \mathbf{E}_D R_D(\ell)$.*

Proof. Let

$$\chi_a^* = \begin{cases} 1 & \text{if } v_a \geq x^* \\ 0 & \text{if } v_a < x^* \end{cases} .$$

At a given point of the sample space, let ℓ denote the outcome leaf, and let $a_1, a_2, \ldots, a_n$ be the ancestors of ℓ. Then the revenue of $\mathcal{S}^*$ is $\sum_{i=1}^n \chi_{a_i}^* x^*$, and the revenue of $\mathcal{S}$ is $\sum_{i=1}^n \chi_{a_i} x_{a_i}$. It follows that the expected difference between the two is

$$\begin{aligned} &\sum_{a \in T} p_D(a)[\mathbf{E}_D(\chi_a^* x^*) - \mathbf{E}_D(\chi_a x_a)] \\ &= \sum_{a \in T} p_D(a)[x^* D(x^*) - x_a D(x_a)] \\ &= \sum_{a \in T} \sum_{\ell \succ a} p_D(\ell) r_D(x_a) \\ &= \sum_{\ell \in T} p_D(\ell) \left(\sum_{a \prec \ell} r_D(x_a) \right) \\ &= \mathbf{E}_D R_D(\ell). \end{aligned}$$

□

3.2.5 Proof of the lower bound

The lemmas in this section will introduce constants $c_1, c_2, \ldots$ which are determined by properties of the demand-curve family $\mathcal{D}$ specified in section 3.2.1. Each such lemma should be interpreted as asserting the existence of a positive, finite constant such that the specified inequality holds.

We begin by establishing that $\mathbf{E}_D K_D$ is bounded above by a constant multiple of $\mathbf{E}_D R_D$. Assume for now that D is fixed, so x^* is also fixed, and put

$$h_a = x_a - x^*.$$

Lemma 3.3. $\mathbf{E}_D R_D(\ell) \geq c_1 \sum_{a \in T} p_D(a) h_a^2$.

Proof. Recall from Lemma 3.1 that

$$(x^* + h) D(x^* + h) < x^* D(x^*) - \beta h^2,$$

hence

$$r_D(x_a) = x^* D(x^*) - x_a D(x_a) > \beta h_a^2.$$

Now we see that

$$\begin{aligned} \mathbf{E}_D R_D(\ell) &= \sum_{\ell \in T} p_D(\ell) \left(\sum_{a \prec \ell} r_D(x_a) \right) \\ &= \sum_{a \in T} \left(\sum_{\ell \succ a} p_D(\ell) \right) r_D(x_a) \\ &= \sum_{a \in T} p_D(a) r_D(x_a) \\ &> \beta \sum_{a \in T} p_D(a) h_a^2. \end{aligned}$$

so the lemma holds with $c_1 = \beta$. □

Lemma 3.4. $\mathbf{E}_D K_D(\ell) \leq c_2 \sum_{a \in T} p_D(a) h_a^2$.

Proof sketch. As in the preceding lemma, the idea is to rewrite the sum over leaves as a sum over internal nodes and then bound the sum term-by-term. A complication arises from the fact that the natural expression for $\mathbf{E}_D K_D(\ell)$ involves summing over pairs of ancestors of a leaf, since $K_D(\ell)$ is the square of $IRE_D(\ell)$, which is the sum of $ire_D(x_a, \chi_a)$ over ancestors of ℓ. This means $\mathbf{E}_D K_D(\ell)$ is expressible as a double sum over pairs of internal nodes of T. However, it turns out that all of the cross-terms cancel, and the remaining terms may all be bounded using part 3 of Lemma 3.1. □

Corollary 3.5. $\mathbf{E}_D K_D(\ell) \leq c_3 \mathbf{E}_D R_D(\ell)$.

The relevance of Corollary 3.5 is that it means that when $\mathbf{E}_D R_D$ is small, then $p_{D_t}(\ell)$ cannot shrink very rapidly as a function of t, for most leaves ℓ. This is made precise by the following Lemma. Here and throughout the rest of this section, D refers to a demand curve $D_{t_0} \in \mathcal{D}$.

Lemma 3.6. *For all sufficiently large n, if $\mathbf{E}_D R_D < \sqrt{n}$ then there exists a set S of leaves such that $p_D(S) \geq 1/2$, and $p_{D_t}(\ell) > c_4 p_D(\ell)$ for all $\ell \in S$ and all $t \in \left[t_0, t_0 + n^{-1/4}\right]$.*

Proof sketch. We must prove that

$$\log p_D(\ell) - \log p_{D_t}(\ell) < -\log c_4$$

for all $\ell \in S$. To do so, we may use the Taylor expansion for the function $g(t) = \log p_{D_t}(\ell)$, truncated at the cubic term. One can prove that the remainder term is $O(1)$ at every leaf, and that at all but a small fraction of leaves (as measured by p_D), all other terms are also $O(1)$. The key ingredients are the upper bound on knowledge coming from Corollary 3.5, the Azuma-Hoeffding martingale inequality, and part 4 of Lemma 3.1. □

We will also need a lemma establishing the growth rate of $R_{D_t}(\ell)$ for a fixed leaf ℓ, as t varies.

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Lemma 3.7. *$R_D(\ell)+R_{D_t}(\ell) > c_5(t-t_0)^2 n$ for all leaves $\ell \in T_n$ and for all $D_t \in \mathcal{D}$.*

Proof. We know that

$$\begin{aligned} R_D(\ell) &= \sum_{a \prec \ell} r_D(x_a) \\ R_{D_t}(\ell) &= \sum_{a \prec \ell} r_{D_t}(x_a) \end{aligned}$$

so it suffices to prove that $r_D(x) + r_{D_t}(x) > c_5(t-t_0)^2$ for all $x \in [0,1]$. Assume without loss of generality that $t - t_0 > 0$. (Otherwise, we may reverse the roles of D and D_t.) Let x^* and x_t^* denote the optimal prices for D, D_t, respectively. (Recall that $D = D_{t_0}$.) Note that $x_t^* - x^* > \alpha(t-t_0)$, by property 1 of Lemma 3.1.

Let $h = x - x^*$, $h_t = x - x_t^*$, and note that $|h| + |h_t| > \alpha(t-t_0)$. Now

$$\begin{aligned} r_D(x) &> c_1|h|^2 \\ r_{D_t}(x) &> c_1|h_t|^2 \\ r_D(x) + r_{D_t}(x) &> c_1(|h|^2 + |h_t|^2) \\ &\geq \frac{1}{2}c_1(|h| + |h_t|)^2 \\ &> \frac{1}{2}c_1\alpha^2 t^2, \end{aligned}$$

so the lemma holds with $c_5 = \frac{1}{2}c_1\alpha^2$. □

We now exploit Lemmas 3.6 and 3.7 to prove that if $\mathbf{E}_D R_D$ is less than some small constant multiple of $\sqrt{n}$, then $\mathbf{E}_{D_t} R_{D_t} = \Omega(\sqrt{n})$ on a large fraction of the interval $[t_0, t_0 + n^{-1/4}]$. The idea behind the proof is that Lemma 3.6 tells us there is a large set S of leaves whose measure does not vary by more than a constant factor as we move t across this interval, while Lemma 3.7 tells us that the regret contribution from leaves in S is $\Omega(\sqrt{n})$ for a large fraction of the t-values in this interval.

In the following proposition, $c(M)$ denotes the function $\min\left\{1, \frac{1}{2}c_4c_5(1+c_4)^{-1}M^{-2}\right\}$.

Proposition 3.8. *For all M and all sufficiently large n, if $\mathbf{E}_D R_D < c(M)\sqrt{n}$, then $\mathbf{E}_{D_t} R_{D_t} > c(M)\sqrt{n}$ for all $t \in \left[t_0 + (1/M)n^{-1/4},\, t_0 + n^{-1/4}\right]$.*

Proof. If $\mathbf{E}_D R_D < c(M)\sqrt{n}$, we may apply Lemma 3.6 to produce a set S of leaves such that $p_D(S) \geq 1/2$ and $p_{D_t}(\ell) > c_4 p_D(\ell)$ for all $\ell \in S$ and all $t \in [t_0, t_0 + n^{-1/4}]$. Now,

$$\mathbf{E}_D R_D \geq \sum_{\ell \in S} p_D(\ell) R_D(\ell)$$

and, for all $t \in [t_0 + (1/M)n^{-1/4}, t_0 + n^{-1/4}]$,

$$\begin{aligned} \mathbf{E}_{D_t} R_{D_t} &\geq \sum_{\ell \in S} p_{D_t}(\ell) R_{D_t}(\ell) \\ &\geq c_4 \sum_{\ell \in S} p_D(\ell) R_{D_t}(\ell) \\ &> c_4 \sum_{\ell \in S} p_D(\ell)(c_5(t-t_0)^2 n - R_D(\ell)) \\ &> c_4 p_D(S) c_5 \sqrt{n}/M^2 - c_4 \mathbf{E}_D R_D \\ &> (c_4c_5/2M^2)\sqrt{n} - c_4 c(M)\sqrt{n} \\ &\geq c(M)\sqrt{n} \end{aligned}$$

where the fourth line is derived from the third by applying the inequality $R_{D_t}(\ell) > c_4(t-t_0)^2 n - R_D(\ell)$ coming from Lemma 3.7. □

Theorem 3.9. *Let $\mathcal{S}$ be any randomized non-uniform strategy, and let $\mathcal{R}_D(\mathcal{S}, n)$ denote the expected ex ante regret of S on a population of n buyers whose valuations are independent random samples from the probability distribution specified by the demand curve D. Then*

$$\Pr_{D \leftarrow \mathcal{D}}\left(\limsup_{n\to\infty} \frac{\mathcal{R}_D(\mathcal{S}, n)}{\sqrt{n}} > 0\right) = 1.$$

In other words, if D is drawn at random from $\mathcal{D}$, then almost surely $\mathcal{R}_D(\mathcal{S}, n)$ is not $o(\sqrt{n})$.

Proof. It suffices to prove the theorem for a deterministic strategy $\mathcal{S}$, since any randomized strategy is a probability distribution over such strategies. Now assume, to the contrary, that

$$\Pr_{D \leftarrow \mathcal{D}}\left(\limsup_{n\to\infty} \frac{\mathcal{R}_D(\mathcal{S}, n)}{\sqrt{n}} = 0\right) > 0. \tag{1}$$

and choose M large enough that the left side of (1) is greater than $1/M$. Recall from Lemma 3.3 that $\mathbf{E}_D R_D = \mathcal{R}_D(\mathcal{S}, n)$. We know that for every $D = D_{t_0} \in \mathcal{D}$ such that $\mathbf{E}_D R_D < c(M)\sqrt{n}$,

$$\mathbf{E}_{D_t} R_{D_t} > c(M)\sqrt{n} \quad \forall t \in [t_0 + (1/M)n^{-1/4}, t_0 + n^{-1/4}]. \tag{2}$$

Now choose N large enough that the set

$$X_N = \left\{D \in \mathcal{D} : \sup_{n > N} \frac{\mathcal{R}_D(\mathcal{S}, n)}{\sqrt{n}} < c(M)\right\}$$

has measure greater than $1/M$. Replacing X_N if necessary with a proper subset still having measure greater than $1/M$, we may assume that $\{t : D_t \in X_N\}$ is disjoint from $[0.4 - \varepsilon, 0.4]$ for some $\varepsilon > 0$. Choosing n large enough that $n > N$ and $n^{-1/4} < \varepsilon$, equation (2) ensures that the sets

$$X_N^k = \{D_s : s = t + (k/M)n^{-1/4}, D_t \in X_N\}$$

are disjoint for $k = 0, 1, \ldots, M-1$. But each of the sets X_N^k, being a translate of X_N, has measure greater than $1/M$. Thus their total measure is greater than 1, contradicting the fact that $\mathcal{D}$ has measure 1. □

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3.2.6 General demand-curve models

The methods of the preceding section extend to much more general families of demand curves. Here we will merely sketch the ideas underlying the extension. Suppose that $\mathcal{D}$ is a compact subset of the space $C^4([0,1])$ of functions on $[0,1]$ with continuous fourth derivative, and that the demand curves $D \in \mathcal{D}$ satisfy the following two additional hypotheses:

- **(Unique global max)** The function $f(x) = xD(x)$ has a unique global maximum $x^* \in [0,1]$, and it lies in the interior of the interval.
- **(Non-degeneracy)** The second derivative of f is strictly negative at x^*.

Suppose $\mathcal{D}$ is also endowed with a probability measure, denoted μ. The proof of the lower bound relied heavily on the notion of being able to make a "one-parameter family of perturbations" to a demand curve. This notion may be encapsulated using a *flow* $\phi(D,t)$ mapping an open set $U \subseteq \mathcal{D} \times \mathbb{R}$ into $\mathcal{D}$, such that $(D \times \{0\}) \cap U$ has measure 1, and $\phi(D,0) = D$ when defined. We will use the shorthand D_t for $\phi(D,t)$. The flow must satisfy the following properties:

- **(Additivity)** $\phi(D, s+t) = \phi(\phi(D,s),t)$.
- **(Measure-preservation)** If $X \subseteq \mathcal{D}$ and $\phi(D,t)$ is defined for all $D \in X$, then $\mu(\phi(X,t)) = \mu(X)$.
- **(Smoothness)** The function $g(t,x) = D_t(x)$ is a C^4 function of t and x.
- **(Profit-preservation)** If x_t^* denotes the point at which the function $xD_t(x)$ achieves its global maximum, then $x_t^* D_t(x_t^*) = x_0^* D_0(x_0^*)$ for all t such that D_t is defined.
- **(Non-degeneracy)** $\frac{d}{dt}(x_t^*) \neq 0$.
- **(Rate dampening at 0 and 1)** For $k = 1,2,3,4$, the functions $\left|\frac{D^{(k)}}{D}\right|$ *and* $\left|\frac{D^{(k)}}{1-D}\right|$ are uniformly bounded above, where $D^{(k)}$ denotes the k-th derivative of D with respect to t.

Provided that these axioms are satisfied, it is possible to establish all of the properties specified in Lemma 3.1. Property 1 follows from compactness of $\mathcal{D}$ and non-degeneracy of ϕ, property 2 follows from the compactness of $\mathcal{D}$ together with the non-degeneracy and "unique global max" axioms for $\mathcal{D}$, and property 4 is the rate-dampening axiom. Property 3 is the subtlest: it follows from the smoothness, profit-preservation, and rate-dampening properties of ϕ. The key observation is that profit-preservation implies that

$$x^* D_t(x^*) \leq x_t^* D_t(x_t^*) = x^* D(x^*),$$

so that $x^* D_t(x^*)$, as a function of t, is maximized at $t = 0$. This, coupled with smoothness of ϕ, proves that $\dot{D}(x^*) = 0$. Another application of smoothness yields the desired bounds.

The final steps of Theorem 1 used the translation-invariance of Lebesgue measure on the interval $[0.3, 0.4]$ to produce M sets whose disjointness yielded the desired contradiction. This argument generalizes, with the flow ϕ playing the role of the group of translations. It is for this reason that we require ϕ to satisfy the additivity and measure-preservation axioms.

3.3 Upper bound

The upper bound on regret in the random-valuation model is based on applying techniques from the literature on the multi-armed bandit problem, specifically [2]. To do so, we discretize the set of possible actions by limiting the seller to strategies which only offer prices belonging to the set $\{1/K, 2/K, \ldots, 1-1/K, 1\}$, for suitably-chosen K. (It will turn out that $K = \theta((n/\log n)^{1/4})$ is the best choice.)

We are now in a setting where the seller must choose one of K possible actions on each of n trials, where each action yields a reward which is a random variable taking values in $[0,1]$, whose distribution depends on the action chosen, but the rewards for a given action are i.i.d. across the n trials. This is the scenario studied in [2]. They define μ^i to be the expected reward of action i, $\mu^* = \max\{\mu_1, \ldots, \mu_K\}$, and

$$\Delta_i = \mu^* - \mu_i. \tag{3}$$

Having made these definitions, the following theorem is proven.

Theorem 3.10 ([2], Theorem 1.). *There exists a strategy* UCB1 *such that, for all $K > 1$, if* UCB1 *is run on a set of K actions having arbitrary reward distributions $P_1, \ldots, P_K$ with support in $[0,1]$, then its expected regret after any number n of plays is at most*

$$\left[8 \sum_{i:\mu_i<\mu^*} \left(\frac{\log n}{\Delta_i}\right)\right] + \left(1 + \frac{\pi^2}{3}\right)\left(\sum_{j=1}^{K} \Delta_j\right).$$

To be precise, the model in [2] assumes that the reward variables for different actions are independent, an assumption which does not hold in our scenario. However, this assumption is not used in their proof of Theorem 3.10, so we may still apply the theorem. To do so, we need to know something about the values of $\Delta_1, \ldots, \Delta_K$ in the special case of interest to us. When the buyer's valuation is v, the payoff of action i/K is

$$X_i = \begin{cases} i/K & \text{if } v \geq i/K \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

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Hence

$$\mu_i = \mathbf{E}(X_i) = (i/K)D(i/K). \tag{5}$$

Recall that we are assuming that the function $f(x) = xD(x)$ has a unique global maximum at $x^* \in (0,1)$, and that $f''(x^*)$ is defined and strictly negative. This hypothesis is useful because it enables us to establish the following lemma, which translates directly into bounds on Δ_i.

Lemma 3.11. *There exist constants C_1, C_2 such that $C_1(x^* - x)^2 < f(x^*) - f(x) < C_2(x^* - x)^2$ for all $x \in [0,1]$.*

Proof. The existence and strict negativity of $f''(x^*)$ guarantee that there are constants $A_1, A_2, \varepsilon > 0$ such that $A_1(x^* - x)^2 < f(x^*) - f(x) < A_2(x^* - x)^2$ for all $x \in (x^* - \varepsilon, x^* + \varepsilon)$. The compactness of $X = \{x \in [0,1] : |x^* - x| \geq \varepsilon\}$, together with the fact that $f(x^*) - f(x)$ is strictly positive for all $x \in X$, guarantees that there are constants B_1, B_2 such that $B_1(x^* - x)^2 < f(x^*) - f(x) < B_2(x^* - x)^2$ for all $x \in X$. Now put $C_1 = \min\{A_1, B_1\}$ and $C_2 = \max\{A_2, B_2\}$ to obtain the lemma. □

Corollary 3.12. *$\Delta_i \geq C_1(x^* - i/K)^2$ for all i. If $\tilde{\Delta}_0 \leq \tilde{\Delta}_1 \leq \ldots \leq \tilde{\Delta}_{K-1}$ are the elements of the set $\{\Delta_1, \ldots, \Delta_k\}$ sorted in ascending order, then $\tilde{\Delta}_j \geq C_1(j/2K)^2$.*

Proof. The inequality $\Delta_i \geq C_1(x^* - i/K)^2$ is a restatement of the lemma using the formulae for Δ_i, μ_i given above in (3),(5). The lower bound on $\tilde{\Delta}_j$ follows upon observing that at most j elements of the set $\{1/K, 2/K, \ldots, 1\}$ lie within a distance $j/2K$ of x^*. □

Corollary 3.13. $\mu^* > x^* D(x^*) - C_2/K^2$.

Proof. At least one of the numbers $\{1/K, 2/K, \ldots, 1\}$ lies within $1/K$ of x^*; now apply the upper bound on $f(x^*) - f(x)$ stated in the lemma. □

Putting all of this together, we obtain the following upper bound.

Theorem 3.14. *Assuming that the function $f(x) = xD(x)$ has a unique global maximum $x^* \in (0,1)$, and that $f''(x^*)$ is defined and strictly negative, the strategy* UCB1 *with $K = \lceil (n/\log n)^{1/4} \rceil$ achieves ex post regret $O(\sqrt{n \log n})$.*

Proof. Consider the following four strategies: UCB1, the strategy defined in [2]; $\mathcal{S}^{\text{opt}}$, the optimal fixed-price strategy; $\mathcal{S}^*$, the fixed-price strategy which offers x^* to every buyer; and $\mathcal{S}^*_K$, the fixed-price strategy which offers i^*/K to every buyer, where i^*/K is the element of $\{1/K, 2/K, \ldots, 1\}$ closest to x^*. As usual, we will use $\rho(\cdot)$ to denote the expected revenue obtained by a strategy. We will prove a $O(\sqrt{n \log n})$ upper bound on each of $\rho(\mathcal{S}^*_K) - \rho(\text{UCB1})$, $\rho(\mathcal{S}^*) - \rho(\mathcal{S}^*_K)$, and $\rho(\mathcal{S}^{\text{opt}}) - \rho(\mathcal{S}^*)$, thereby establishing the theorem.

The fact that $\rho(\mathcal{S}^*_K) - \rho(\text{UCB1}) = O(\sqrt{n \log n})$ follows from Theorem 3.10 and Corollary 3.12, while the bound $\rho(\mathcal{S}^*) - \rho(\mathcal{S}^*_K) = O(\sqrt{n \log n})$ is an immediate consequence of Corollary 3.13.

It remains to bound $\rho(\mathcal{S}^{\text{opt}}) - \rho(\mathcal{S}^*)$. For any $x \in [0,1]$, let $\rho(x)$ denote the revenue obtained by the fixed-price strategy which offers price x, and let $x^{\text{opt}} = \arg\max_{x \in [0,1]} \rho(x)$. We begin by observing that for all $x < x^{\text{opt}}$,

$$\rho(x) \geq \rho(x^{\text{opt}}) - n(x^{\text{opt}} - x).$$

This is simply because every buyer that accepts price x^{opt} would also accept x, and the amount of revenue lost by setting the lower price is $x^{\text{opt}} - x$ per buyer. Now

$$\begin{aligned}
&\int_0^1 \Pr(\rho(x) - \rho(x^*) > \lambda)\, dx \\
&\quad \geq \int_0^1 \Pr(\rho(x^{\text{opt}}) - \rho(x^*) > 2\lambda \text{ and } x^{\text{opt}} - x < \lambda/n)\, dx \\
&\quad = (\lambda/n) \Pr(\rho(x^{\text{opt}}) - \rho(x^*) > 2\lambda),
\end{aligned}$$

so a bound on $\Pr(\rho(x) - \rho(x^*) > \lambda)$ for each fixed x translates into a bound on $\Pr(\rho(x^{\text{opt}}) - \rho(x^*) > \lambda)$. But for fixed x, the probability in question is the probability that a sum of n i.i.d. random variables, each supported in $[-1,1]$ and with negative expectation, exceeds λ. The Chernoff-Hoeffding bound tells us that $\Pr(\rho(x) - \rho(x^*) > \lambda) < e^{-\lambda^2/2n}$, so

$$\Pr(\rho(x^{\text{opt}}) - \rho(x^*) > 2\lambda) < \min\{1, \frac{n}{\lambda} e^{-\lambda^2/2n}\}.$$

Finally,

$$\begin{aligned}
\mathbf{E}(\rho(x^{\text{opt}}) - \rho(x^*)) &< \int_0^\infty \Pr(\rho(x^{\text{opt}}) - \rho(x^*) > y)\, dy \\
&< \int_0^\infty \min\{1, \frac{2n}{y} e^{-y^2/8n}\}\, dy \\
&< \int_0^{\sqrt{4n \log n}} dy \\
&\quad + \frac{2n}{\sqrt{4n \log n}} \int_{\sqrt{4n \log n}}^\infty e^{-y^2/8n} dy \\
&= O(\sqrt{n \log n}).
\end{aligned}$$

□

Remark 3.15. If the seller does not have foreknowledge of n, it is still possible to achieve regret $O(\sqrt{n \log n})$ by maintaining an estimate n_{est} of n, initialized to 1. When 2^k buyers have been seen, the seller sets n_{est} to 2^{k+1} and reinitializes UCB1 using this new value of n_{est}.

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4 Worst-case valuations

In the worst-case valuation model, we assume that the buyers' valuations are chosen by an adversary who has knowledge of n and of the pricing strategy, but is oblivious to the algorithm's random choices. A similar adversarial model for the multi-armed bandit problem was considered by Auer et al in [3]. In that paper the authors present an algorithm **Exp3** achieving regret $O(\sqrt{nK \log K})$, where K is the number of possible actions, and they exhibit a lower bound of $\Omega(\sqrt{nK})$ on regret. The algorithm, which is based on the weighted-majority learning algorithm of Littlestone and Warmuth, was applied in the setting of on-line auctions by Blum et al [6], who normalize the buyers' valuations to lie in an interval $[1, h]$ and then prove the following theorem:

Theorem 4.1 ([6], Theorem 5.). *There exists a pricing strategy* **Exp3** *and a constant $c(\varepsilon)$ such that for all valuation sequences, if the optimal fixed-price revenue $\rho(\mathcal{S}^{\mathrm{opt}})$ satisfies $\rho(\mathcal{S}^{\mathrm{opt}}) > c(\varepsilon) h \log h \log\log h$, then* **Exp3** *is $(1+\varepsilon)$-competitive relative to $\rho(\mathcal{S}^{\mathrm{opt}})$.*

Our upper and lower bounds for regret in the worst-case valuation model are based on the techniques employed in these two papers. The upper bound asserted in Theorem 1.3 is virtually a restatement of Blum et al's theorem, though the change in emphasis from competitive ratio to additive regret necessitates a minor change in technical details. Our worst-case lower bound (Theorem 4.3) is influenced by Auer et al's proof of the corresponding lower bound for the adversarial multi-armed bandit problem in [3].

4.1 Upper bound

Following [6], as well as the technique used in Section 3.3 above, we specify a finite set of offer prices $X = \{1/K, 2/K, \ldots, 1\}$ and constrain the seller to select prices from this set only. This reduces the posted-price auction problem to an instance of the multi-armed bandit problem, to which the algorithm **Exp3** of [3] may be applied. Denote this pricing strategy by $\mathcal{S}$. The relevant theorem about **Exp3** is the following.

Theorem 4.2 ([3], Corollary 4.2.). *If one runs the algorithm* **Exp3** *with a set of K actions, over n steps, with the rewards for each action in each step belonging to $[0, 1]$, then the expected regret of* **Exp3** *relative to the best fixed action is at most $2\sqrt{e-1}\sqrt{nK \log K}$.*

Thus, if $\mathcal{S}_K^{\mathrm{opt}}$ denotes the fixed-price strategy which chooses the best offer price i^*/K from X, and $\mathcal{S}^{\mathrm{opt}}$ denotes the fixed-price strategy which chooses the best offer price x^{opt} from $[0, 1]$, we have the following inequalities:

$$\rho(\mathcal{S}_K^{\mathrm{opt}}) - \rho(\mathcal{S}) \quad < \quad 2\sqrt{e-1}\sqrt{nK \log K}$$

$$\rho(\mathcal{S}^{\mathrm{opt}}) - \rho(\mathcal{S}_K^{\mathrm{opt}}) \quad < \quad n(1/K) = n/K$$

where the second inequality follows from the fact that $\mathcal{S}_K^{\mathrm{opt}}$ is no worse than the strategy which offers $\frac{1}{K}\lfloor K x^{\mathrm{opt}} \rfloor$ to each buyer.

If we pick $K = \lceil n/\log n \rceil^{1/3}$, then both $\sqrt{nK \log K}$ and n/K are $O(n^{2/3}(\log n)^{1/3})$. We have thus expressed the regret of **Exp3** as a sum of two terms, each of which is $O(n^{2/3}(\log n)^{1/3})$, establishing the upper bound asserted in Theorem 1.3. When the seller does not have foreknowledge of n, the upper bound may be achieved using the same doubling argument given in Remark 3.15.

Readers familiar with [6] will recognize that the only difference between this argument and their Theorem 5 is that they choose the prices in X to form a geometric progress (so as to optimize the competitive ratio) while we choose them to form an arithmetic progression (so as to optimize the additive regret).

4.2 Lower bound

In [3], the authors present a lower bound of $\sqrt{nK}$ for the multi-armed bandit problem with payoffs selected by an oblivious adversary. The authors define a distribution of random payoffs (depending on n) such that the expected regret of *any* algorithm on a random sample from this distribution is $\Omega(\sqrt{nK})$. The idea is select one of the K actions uniformly at random and designate it as the "good" action. For all other actions, the payoff in each round is a uniform random sample from $\{0, 1\}$, but for the good action the payoff is 1 with probability $1/2 + \varepsilon$, 0 otherwise. Here $\varepsilon = \theta(\sqrt{K/n})$. A strategy which knows the good action will achieve expected payoff $(1/2+\varepsilon)n = 1/2+\theta(\sqrt{nK})$. It can be shown, for information-theoretic reasons, that no strategy can learn the good action rapidly enough to play it more than $n/K + \theta(\varepsilon\sqrt{n^3/K})$ times in expectation, from which the lower bound on regret follows.

A similar counterexample can be constructed in the context of our on-line posted-price auction problem, yielding expected regret $\Omega(n^{2/3})$. The idea is roughly the same as above: one randomly chooses a subinterval of $[0, 1]$ of length $1/K$ to be the interval of "good prices," and chooses the distribution of buyers' valuations so that the expected revenue per buyer is a constant independent of the offer price outside the interval of good prices, and is ε higher than this constant inside the interval of good prices. As above, there is a trade-off between choosing ε too large (which makes it too easy for strategies to learn which prices belong to the good interval) or too small (which leads to a negligible difference in revenue between the best strategy and all others), and the optimal trade-off is achieved when $\varepsilon = \theta(\sqrt{K/n})$. However, in our setting there is an additional constraint that $\varepsilon \leq 1/K$, since the expected payoff

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can grow by no more than $1/K$ on an interval of length $1/K$. This leads to the values $K = \theta(n^{1/3}), \varepsilon = \theta(n^{-1/3})$ and yields the stated lower bound of $\Omega(n^{2/3})$.

Theorem 4.3. *For any given n, there exists a finite family $\mathcal{P} = \{p_j^n\}_{j=1}^K$ of probability distributions on $[0,1]$, such that if p_j^n is chosen uniformly at random from $\mathcal{P}$ and then buyers' valuations are sampled independently at random according to p_j^n, no pricing strategy can achieve expected regret $o(n^{2/3})$, where the expectation is over both the random choice of D and the randomly-sampled valuations.*

Proof sketch. For simplicity, assume $n = 8K^3$, and put $\varepsilon = 1/2K$. The valuations will be independent random samples from the set $V = \{\frac{1}{2}, \frac{1}{2}+\varepsilon, \frac{1}{2}+2\varepsilon, \ldots, 1-\varepsilon, 1\}$. A "baseline probability distribution" p_{base} on V is defined so that

$$p_{\text{base}}(\{v \geq 1 - i\varepsilon\}) = \frac{1}{2}(1 - i\varepsilon)^{-1}.$$

A random sample from $p_j^n \in \mathcal{P}$ is generated by sampling $v \in V$ at random from the distribution p_{base}, and then adding ε to it with probability 1/10 if and only if $v = 1 - j\varepsilon$.

If the adversary selects a random p_j^n from $\mathcal{P}$, then the expected regret of any pricing strategy $\mathcal{S}$ is $\Omega(n^{2/3})$, where the expectation is over the choice of p_j^n as well as the algorithm's random choices, if any. The proof if this fact exactly parallels the proof of the lower bound in [3] (Theorem 6.1). There are two complications that come up along the way. One is that the seller's algorithm has a continuum of alternatives at every step, rather than a finite set of K alternatives. This can be dealt with by restricting the buyers' valuations to lie in a finite set $V = \{v_1, v_2, \ldots, v_K\}$. Then there is no incentive for the seller to offer a price which lies outside of V, so we may assume the seller is constrained to offer prices in V and prove lower bounds for this restricted class of strategies.

The second complication is that the adversary in [3] was more powerful: he could specify the reward for each action independently, whereas our adversary can only set a valuation v, and this v determines the rewards for all actions simultaneously. While this entails choosing a more complicated reward distribution, the complication only makes the computations messier without introducing any new ideas into the proof. □

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